

Introduction

- Flowlines are underground pipes transporting oil, gas, and water.
- Failures can cause environmental harm and human risk.
- Objective:** Develop predictive models to identify high-risk flowlines in Colorado using real-world regulatory data.

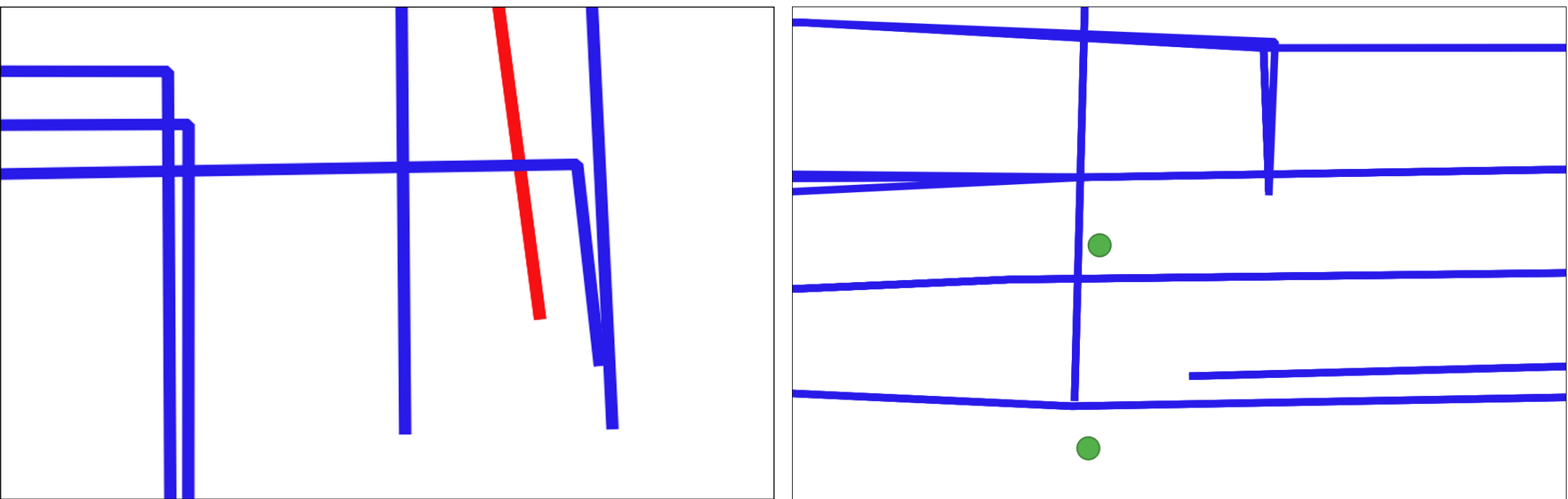
Data Processing

Data Sources:

- 1,726 ECMC-reported spill events (2014–2022) [4]
- 21,000+ flowlines (spatial and operational attributes)

Spatial Matching:

- Operational and descriptive datasets merged by endpoints using a 25m tolerance.
- 4,117 flowlines were successfully matched.
- 41 spills were linked to specific flowlines with high confidence.



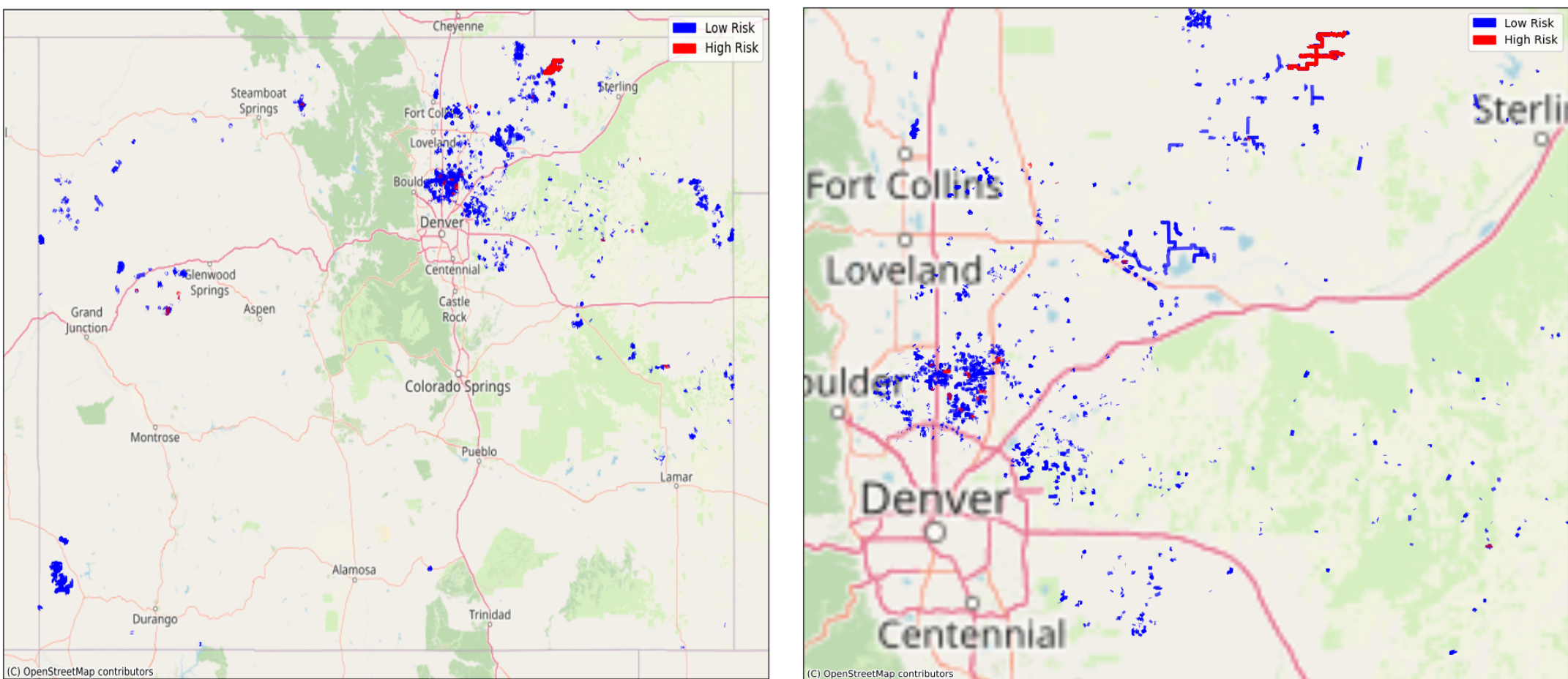
Feature Engineering:

- Created binary **risk** variable: 1 = spill, 0 = no spill.
- Spatial features extracted: line complexity, bounding box area, etc.
- Categorical variables one-hot encoded (e.g., material, fluid type).

Attributes for Flowline Analysis

Attribute	Description	Units
Status	Operational status of the flowline	-
Flowline Action	Actions taken or required on the flowline	-
Location Type	Type of facility	-
Fluid Type	Type of fluid transported	-
Material	Construction material of the flowline	-
Diameter	Diameter of the flowline	Inches
Length	Length of the flowline	Feet
Max Operating Pressure	Max pressure flowline can withstand	PSI
Line Age	Age of the flowline	Years
Number of Lines	Number of line segments (geometry complexity)	-
Bounding Box Area	Area enclosing the flowline	Feet ²
Root Cause Type*	Underlying cause of the spill	-

Observed Data



Supervised ML Models

- Logistic Regression (**LR**): Estimates spill probability using [2]:

$$P(Y = 1 | \mathbf{X}) = \frac{1}{1 + \exp(-z)}$$

- K-Nearest Neighbors (**KNN**): Assigns class based on Euclidean distance:

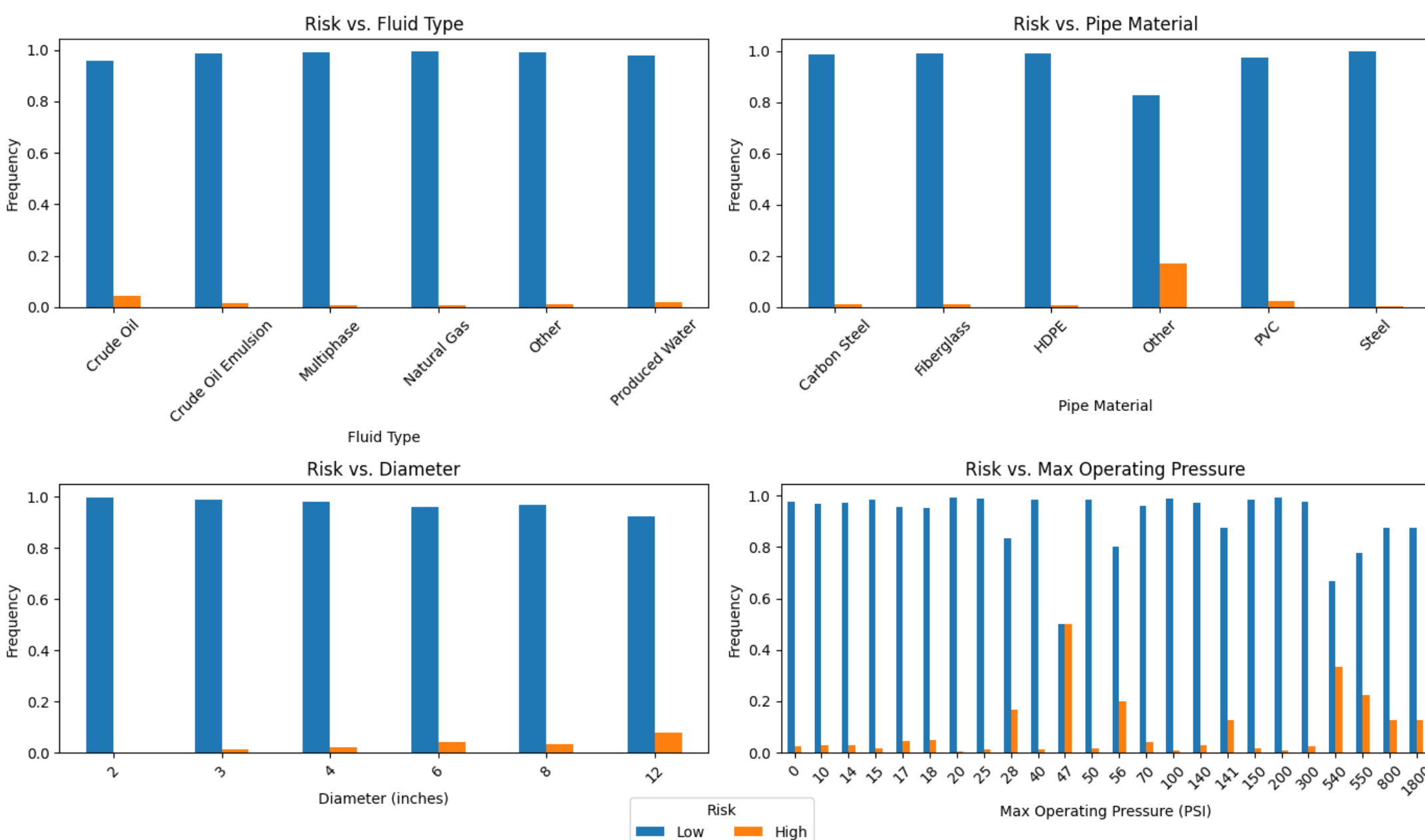
$$d(\mathbf{x}, \mathbf{x}_i) = \sqrt{\sum_{j=1}^n (x_j - x_{i,j})^2}$$

- Support Vector Machine (**SVM**): Finds a hyperplane that maximizes the margin using [2]:

$$\hat{y} = \text{sgn}(\mathbf{w}^T \mathbf{x} + b)$$

- Gradient Boosting (**GBDT**): Trains trees sequentially to reduce prior errors.
- Adaptive Boosting (**AdaBoost**): Reweights data to emphasize hard-to-classify points.
- Random Forest (**RF**): Averages predictions from trees fit on bootstrapped samples. [1]

Risk Assessment of Operational Parameters



Dimensionality Reduction Methods

- Principal Component Analysis (**PCA**): An *unsupervised* method that finds new axes (directions) capturing the most variance in the data. It solves:

$$\max_{\mathbf{w}} \text{Var}(\mathbf{X}\mathbf{w}) \quad \text{s.t. } \|\mathbf{w}\| = 1$$

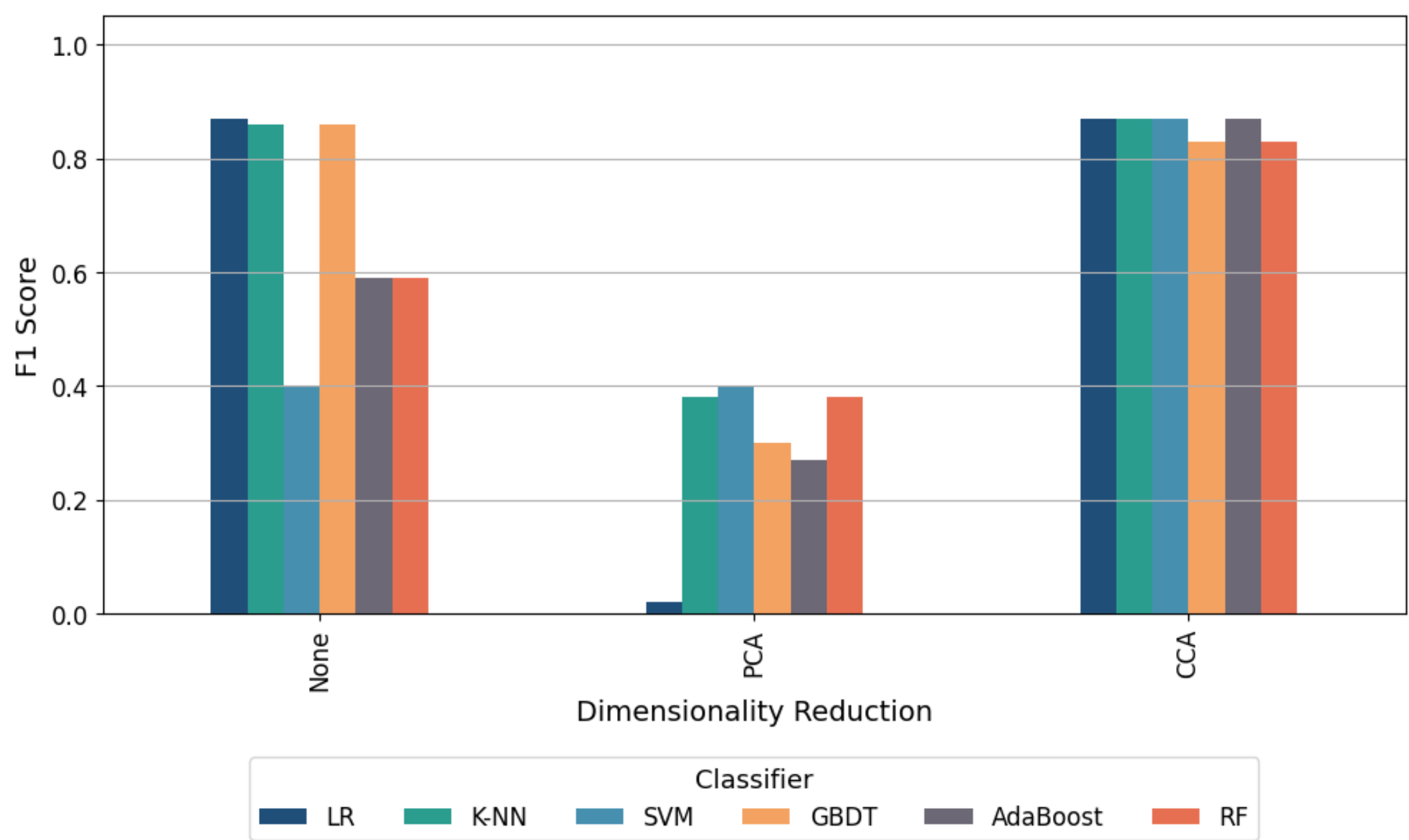
where $\mathbf{w}$ is a direction in feature space. Used to compress data while keeping the most information. [2]

- Canonical Correlation Analysis (**CCA**): A *supervised* method that finds combinations of features most correlated with the target. It solves:

$$\max_{\mathbf{a}} \text{Corr}(\mathbf{X}\mathbf{a}, \mathbf{y})$$

where $\mathbf{X}$ is the feature matrix and $\mathbf{y}$ is the target. Great for focusing on patterns tied to prediction. [3]

F1 Score Across Models and Reduction Methods



Summary & Future Work

We benchmarked six classifiers across three reduction methods to predict spill risk. CCA preserved performance best; PCA reduced it.

- Incorporate spatial autocorrelation into model design using geostatistical methods.
- Expand to time-aware risk predictions and more granular flowline attributes.

References

- [1] L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- [2] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, NY, USA, 2 edition, 2009.
- [3] H. Hotelling. Relations between two sets of variates. *Biometrika*, 28(3/4):321–377, 1936.
- [4] Oil and Gas Conservation Commission, Colorado. Annual flowline spill report – 2019. Technical report, Colorado Oil and Gas Conservation Commission, Denver, CO, USA, 2019.